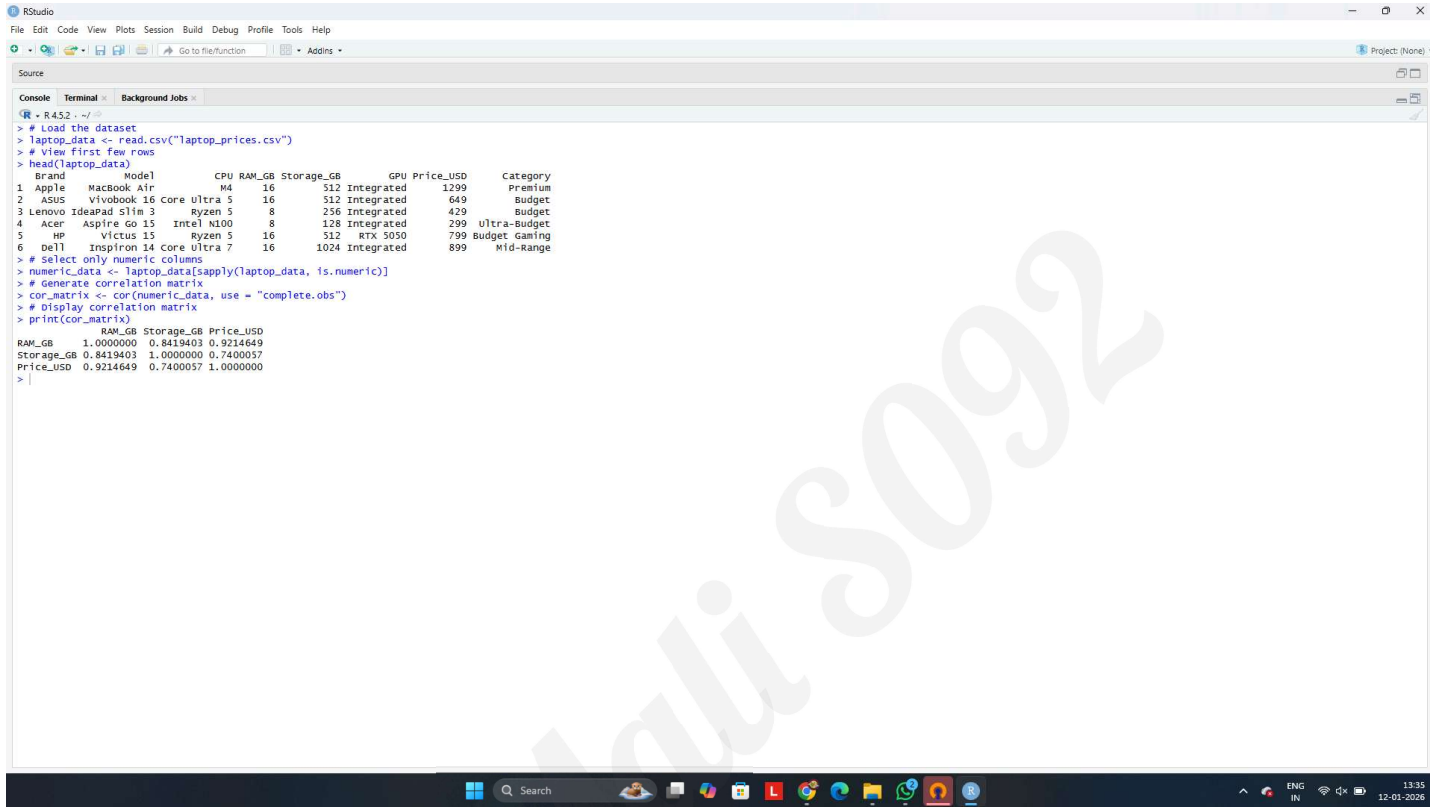


Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis With SAS/SPSS/R
Module 2 Practical 12

Aim :- Generating correlation matrices using cor() (R).



The screenshot shows the RStudio interface with the following code in the console:

```
<R> R 4.5.2 ~ /  
> # Load the dataset  
> laptop_data <- read.csv("laptop_prices.csv")  
> # View first few rows  
> head(laptop_data)  
  Brand      Model      CPU RAM_GB Storage_GB GPU Price_USD Category  
1 Apple  MacBook Air      M4      16        512 Integrated    1299 Premium  
2 ASUS   Vivobook 16 core ultra 5      16        512 Integrated    649 Budget  
3 Lenovo Ideapad Slim 3      Ryzen 5      8        256 Integrated    429 Budget  
4 Acer   Aspire Go 15      Intel N100      8        128 Integrated    299 Ultra-Budget  
5 HP     Victus 15      Ryzen 5      16        512 RTX 5050      799 Budget Gaming  
6 Dell   Inspiron 14 core ultra 7      16       1024 Integrated    899 Mid-range  
> # select only numeric columns  
> numeric_data <- laptop_data[sapply(laptop_data, is.numeric)]  
> # generate correlation matrix  
> cor_matrix <- cor(numeric_data, use = "complete.obs")  
> # Display correlation matrix  
> print(cor_matrix)  
      RAM_GB Storage_GB Price_USD  
RAM_GB  1.0000000  0.8419403  0.9214649  
Storage_GB 0.8419403  1.0000000  0.7400057  
Price_USD  0.9214649  0.7400057  1.0000000  
> |
```

The output shows the correlation matrix for the numeric variables: RAM_GB, Storage_GB, and Price_USD. The matrix is symmetric, with diagonal elements all equal to 1.0000000. The correlation between RAM_GB and Storage_GB is 0.8419403, between RAM_GB and Price_USD is 0.9214649, and between Storage_GB and Price_USD is 0.7400057.

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